

A Packing of 13 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 13 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.888584. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.871484, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 13 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.888584 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 13$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.871484, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 13/12 = 2.888889$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N13.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
515818597178166533841064591764399107893861203038683939081171729827221720
388979063765991167891118784064183587035880448845307375932307118615042178
92956400616568941178847696604138356709228167515
```

denominator:

```
178571428571204257392949083221089042012689099644351527166659653236027465
656991782114934121099718096587789089571727652644330252901671049335702638
40342627724638830646510316816185621825932863904
```

B Coordinates

13 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N13.txt`.

```
64730090956 174210120808 393525869977 -210919285690-146292439726 680648408367 -17163916800 -222719896725063274002 -101146245723 201839441123 413011876399
329683091544 -195704550032 -1806200096 471872615972-104007544271 -253328084238 52298954705 -13165950684323416874656 -242273707235 47531547949 39144528179
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